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Substrate processing method and substrate processing apparatus

Abstract

A substrate processing method is a method of performing an etching process on a substrate with an etching solution in a processing tank. The substrate includes silicon oxide films and silicon nitride films stacked alternately. The etching solution contains phosphoric acid. The substrate processing method includes immersing the substrate in the etching solution, and replenishing the etching solution in the processing tank with phosphoric acid during the etching process on the substrate to cause a silicon concentration in the etching solution to vary.

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Background/Summary

INCORPORATION BY REFERENCE

(1) The present application claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2021-083202, filed on May 17, 2021. The contents of this application are incorporated herein by reference in their entirety.

BACKGROUND

(2) The subject matter of the present application relates to a substrate processing method and a substrate processing apparatus.

(3) There is known a substrate processing apparatus that performs an etching process on a substrate having a multilayer structure in which silicon oxide films and silicon nitride films are alternately stacked. As this kind of substrate processing apparatus, there is a batch type substrate processing apparatus for etching a substrate with etchant containing phosphoric acid. This type of the substrate processing apparatus selectively etches, of silicon oxide films and silicon nitride films, mainly the silicon nitride films, thereby removing the silicon nitride films.

SUMMARY

(4) A substrate processing apparatus that selectively etches mainly silicon nitride films is used for a process by which an etched multilayer structure becomes a structure in which a plurality of flat silicon oxide films are arranged in a comb shape, because the substrate processing apparatus does not substantially etch the silicon oxide films. It is however required to process a substrate into a more complicated shape due to miniaturization and high integration of semiconductor devices.

(5) A substrate processing method according to an aspect of the present disclosure is a method of performing an etching process on a substrate with an etching solution in a processing tank. The substrate includes silicon oxide films and silicon nitride films stacked alternately. The etching solution contains phosphoric acid. The substrate processing method includes immersing the substrate in the etching solution, and replenishing the etching solution in the processing tank with phosphoric acid during the etching process on the substrate to cause a silicon concentration in the etching solution to vary.

(6) In an embodiment, in the replenishing, a replenishment flow rate of the phosphoric acid that the etching solution is replenished with is controlled based on a set value for a replenishment flow rate of phosphoric acid that is set according to a structure of a semiconductor device to be manufactured using the substrate.

(7) In an embodiment, in the replenishing, a replenishment flow rate of the phosphoric acid that the etching solution is replenished with is controlled based a silicon concentration in the etching solution. The silicon concentration is measured during the etching process on the substrate.

(8) In an embodiment, in the replenishing, the phosphoric acid that the etching solution is replenished with is supplied with a silicon containing liquid. The silicon containing liquid is a liquid that contains silicon.

(9) In an embodiment, in the replenishing, a supply flow rate of the silicon containing liquid to be supplied to the phosphoric acid is controlled based on a set value for a supply flow rate of a silicon containing liquid that is set according to a structure of a semiconductor device to be manufactured using the substrate.

(10) In an embodiment, in the replenishing, a supply flow rate of the silicon containing liquid to be supplied to the phosphoric acid is controlled based on a silicon concentration in the etching solution measured during the etching process on the substrate.

(11) In an embodiment, the structure of the semiconductor device indicates size of a gap between the silicon oxide films adjacent to each other in a stacking direction in the semiconductor device.

(12) A substrate processing apparatus according to another aspect of the present disclosure

performs an etching process on a substrate with an etching solution. The substrate includes silicon oxide films and silicon nitride films stacked alternately. The etching solution contains phosphoric acid. The substrate processing apparatus includes a processing tank, a substrate holding section, a phosphoric acid replenishment mechanism, and a controller. The processing tank stores the etching solution. The substrate holding section holds the substrate in the etching solution stored in the processing tank. The phosphoric acid replenishment mechanism replenishes the etching solution in the processing tank with phosphoric acid. The controller controls the phosphoric acid replenishment mechanism during the etching process on the substrate to cause a silicon concentration in the etching solution to vary.

(13) In an embodiment, the controller controls the phosphoric acid replenishment mechanism during the etching process on the substrate based on a set value for a replenishment flow rate of phosphoric acid that is set according to a structure of a semiconductor device to be manufactured using the substrate.

(14) In an embodiment, the substrate processing apparatus further includes a silicon concentration measuring device that measures a silicon concentration in the etching solution. The controller controls the phosphoric acid replenishment mechanism during the etching process on the substrate based on a silicon concentration measured through the silicon concentration measuring device during the etching process on the substrate.

(15) In an embodiment, the substrate processing apparatus further includes a silicon supply mechanism. The silicon supply mechanism supplies a silicon containing liquid to the phosphoric acid that the etching solution is replenished with. The silicon containing liquid is a liquid that contains silicon.

(16) In an embodiment, the controller controls the silicon supply mechanism during the etching process on the substrate based on a set value for a supply flow rate of a silicon containing liquid that is set according to a structure of a semiconductor device to be manufactured using the substrate.

(17) In an embodiment, the substrate processing apparatus further includes a silicon concentration measuring device that measures a silicon concentration in the etching solution. The controller controls the silicon supply mechanism during the etching process on the substrate based on a silicon concentration measured through the silicon concentration measuring device during the etching process on the substrate.

(18) In an embodiment, the structure of the semiconductor device indicates size of a gap between the silicon oxide films adjacent to each other in a stacking direction in the semiconductor device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. **1**A and **1**B illustrate a substrate processing apparatus according to a first embodiment.

(2) FIG. **2** is a cross-sectional view illustrating the configuration of the substrate processing apparatus according to the first embodiment.

(3) FIG. **3** is a diagram illustrating a phosphoric acid replenishment mechanism included in the substrate processing apparatus according to the first embodiment.

(4) FIG. **4** is a flow chart illustrating a substrate processing method according to the first embodiment.

(5) FIG. **5** is a diagram illustrating a varying example of a silicon concentration during an etching process by the substrate process apparatus according to the first embodiment.

(6) FIG. **6** is a diagram illustrating a transitional example of a phosphoric acid replenishment flow rate during the etching process through the substrate processing apparatus according to the first embodiment.

(7) FIG. **7** is a diagram illustrating a substrate before undergoing an etching process through the substrate processing apparatus according to the first embodiment.
(8) FIG. **8** is a diagram illustrating an example of the substrate after undergoing an etching process through the substrate processing apparatus according to the first embodiment.
(9) FIG. **9** is a block diagram illustrating the configuration of a control device included in the substrate processing apparatus according to the first embodiment.
(10) FIG. **10** is a diagram illustrating first to third varying examples of corresponding silicon concentrations during their respective etching processes through the substrate processing apparatus according to the first embodiment.
(11) FIG. **11** is a diagram illustrating first to third transitional examples of corresponding phosphoric acid replenishment flow rates during the respective etching processes through the substrate processing apparatus according to the first embodiment.
(12) FIG. **12** is a diagram illustrating a first example of a substrate after undergoing an etching process through the substrate processing apparatus according to the first embodiment.
(13) FIG. **13** is a diagram illustrating a second example of a substrate after undergoing an etching process through the substrate processing apparatus according to the first embodiment.
(14) FIG. **14** is a diagram illustrating a third example of a substrate after undergoing an etching process through the substrate processing apparatus according to the first embodiment.
(15) FIG. **15** is a diagram illustrating a fourth varying example of the silicon concentration during an etching process by the substrate processing apparatus according to the first embodiment.
(16) FIG. **16** is a diagram illustrating a fourth example of the substrate after undergoing an etching process through the substrate processing apparatus according to the first embodiment.
(17) FIG. **17** is a cross-sectional view illustrating the configuration of a substrate processing apparatus according to a second embodiment.
(18) FIG. **18** is a cross-sectional view illustrating the configuration of a substrate processing apparatus according to a third embodiment.
(19) FIG. **19** is a diagram illustrating a phosphoric acid replenishment mechanism and a silicon supply mechanism included in the substrate processing apparatus according to the third embodiment.
(20) FIG. **20** is a diagram illustrating first to third varying examples of corresponding silicon concentrations during their respective etching processes through the substrate processing apparatus according to the third embodiment.
(21) FIG. **21** is a diagram illustrating first to third transitional examples of corresponding silicon supply flow rates during the respective etching processes through the substrate processing apparatus according to the third embodiment.
(22) FIG. **22** is a diagram illustrating a first example of a substrate after undergoing an etching process through the substrate processing apparatus according to the third embodiment.
(23) FIG. **23** is a diagram illustrating a second example of a substrate after undergoing an etching process through the substrate processing apparatus according to the third embodiment.
(24) FIG. **24** is a diagram illustrating a third example of a substrate after undergoing an etching process through the substrate processing apparatus according to the third embodiment.

DETAILED DESCRIPTION

(25) Hereinafter, embodiments of a substrate processing method and a substrate processing apparatus according to the subject matter of the present application will be described with reference to the drawings (FIGS. **1**A to **24**). However, the subject matter of the present application is not limited to the following embodiments. Note that the duplicated description may be omitted as appropriate. In the drawings, elements which are the same or equivalent are labelled the same reference signs and description thereof is not repeated.
(26) In the present specification, X-direction, Y-direction, and Z-direction orthogonal to each other may be described for ease of understanding. Typically, the X-direction and Y-direction are parallel

to the horizontal direction and the Z-direction is parallel to the vertical direction. However, the definition of these directions does not intend to limit the orientation when the substrate processing method according to the present disclosure is executed and the orientation when the substrate processing apparatus according to the present disclosure is used.

(27) Examples of a "substrate" in the embodiments include a semiconductor wafer, a glass substrate for a photomask, a glass substrate for a liquid crystal display, a glass substrate for a plasma display, a substrate for a field mission display (FED), a substrate for an optical disk, a substrate for a magnetic disk, and a substrate for a magneto-optical disk. Although the embodiments will hereinafter be described mainly by taking as an example a substrate processing method and a substrate processing apparatus used for processing a disk-shaped semiconductor wafer, the embodiments can be similarly applied to the various substrates exemplified above. Further, various shapes can be applied to the shape of the substrate.

First Embodiment

(28) Hereinafter, a first embodiment will be described with reference to FIGS. **1**A to **16**. First, a substrate processing apparatus **100** according to the present embodiment will be described with reference to FIGS. **1**A and **1**B. The substrate processing apparatus **100** according to the present embodiment is a batch type etching apparatus. Therefore, the substrate processing apparatus **100** performs an etching process on a plurality of substrates W at once. For example, the substrate processing apparatus **100** performs an etching process on a plurality of substrates W for each lot unit. One lot consists of, for example 25 substrates W.

(29) FIGS. **1**A and **1**B illustrate the substrate processing apparatus **100** according to the present embodiment. Specifically, FIG. **1**A illustrates the substrate processing apparatus **100** before the substrates W are placed in a processing tank **3**. FIG. **1**B illustrates the substrate processing apparatus **100** after the substrates W are placed in the processing tank **3**. As illustrated in FIGS. **1**A and **1**B, the substrate processing apparatus **100** includes the processing tank **3**, a control device **110**, an elevating section **120**, and a substrate holding portion **130**.

(30) The processing tank **3** stores an etching solution E. The etching solution E contains phosphoric acid (H.sub.3PO.sub.4) and silicon. The etching solution E may further contain a diluting liquid. The diluting liquid is, for example deionized water (DIW). DIW is a kind of pure water. Examples of the diluting liquid include carbonated water, electrolytic ionized water, hydrogen water, ozone water, or hydrochloric acid water having a diluted concentration (for example, about 10 ppm to about 100 ppm). Note that the etching solution E may further contain an additive different from silicon. The processing tank **3** includes an inner tank **31** and an outer tank **32**. The outer tank **32** surrounds the inner tank **31**. In other words, the processing tank **3** has a double tank structure. Each of the inner tank **31** and the outer tank **32** has an upper opening that opens upward.

(31) Each of the inner tank **31** and the outer tank **32** stores the etching solution E. The inner tank **31** allows the plurality of substrates W to be placed in. Specifically, the plurality of substrates W held by the substrate holding portion **130** are placed in the inner tank **31**. The plurality of substrates W are placed in the inner tank **31**, and consequently immersed in the etching solution E in the inner tank **31**.

(32) The substrate holding portion **130** holds the plurality of substrates W in the etching solution E in the processing tank **3** (inner tank **31**). Specifically, the substrate holding portion **130** includes a plurality of holding rods **131** and a body plate **132**. The body plate **132** is a plate-shaped member and extends in the vertical direction (Z-direction). The plurality of holding rods **131** extend in the horizontal direction (Y-direction) from one main surface of the body plate **132**. In the present embodiment, the substrate holding portion **130** includes three holding rods **131** (see FIG. **2**).

(33) The plurality of substrates W are held by the plurality of holding rods **131**. Specifically, the plurality of substrates W are held in an upright posture (vertical posture) by the plurality of holding rods **131** with the lower edges of the substrates W being in contact with the plurality of holding rods **131**. The plurality of substrates W held by the substrate holding portion **130** are arranged at

intervals in the Y-direction. That is, the plurality of substrates W are aligned in a row in the Y-direction. Each of the plurality of substrates W is also held by the substrate holding portion **130** in a posture substantially parallel to the XZ plane.
(34) The control device **110** controls the operation of each section of the substrate processing apparatus **100**. The control device **110** controls, for example the operation of the elevating section **120**. The elevating section **120** is controlled by the control device **110** to lift the substrate holding section **130** up and down. The elevating section **120** lifts the substrate holding section **130** up and down, whereby the substrate holding section **130** moves vertically upward and vertically downward while holding the plurality of substrates W. The elevating section **120** includes a drive source and an elevating mechanism. The elevating mechanism is driven by the drive source to lift the substrate holding section **130** up and down. The drive source includes, for example a motor. The elevating mechanism includes, for example, a rack and pinion mechanism or a ball screw.
(35) More specifically, the elevating section **120** lifts the substrate holding section **130** up and down between a processing position (position in FIG. **1**B) and a retracted position (position in FIG. **1**A). As illustrated in FIG. **1**B, when the substrate holding section **130** moves vertically downward (Z-direction) while holding the plurality of substrates W and reaches the processing position, the plurality of substrates W are placed in the processing tank **3**. That is, the plurality of substrates W held by the substrate holding section **130** move in the inner tank **31**. As a result, the plurality of substrates W are immersed in the etching solution E in the inner tank **31**, and an etching process is performed on the substrates W with the etching solution E. On the other hand, as illustrated in FIG. **1**A, when the substrate holding section **130** moves to the retracted position, the plurality of substrates W held by the substrate holding section **130** move above the processing tank **3** to be pulled up from the etching solution E.
(36) The configuration of the substrate processing apparatus **100** according to the present embodiment will be then described with reference to FIG. **2**. FIG. **2** is a cross-sectional view illustrating the configuration of the substrate processing apparatus **100** according to the present embodiment. As illustrated in FIG. **2**, the control device **110** includes a controller **111** and storage **112**.
(37) The controller **111** may include a processor. Examples of the controller **111** include a central processing unit (CPU) and a micro processing unit (MPU). The controller **111** controls the operation of each section of the substrate processing apparatus **100** based on a computer program and data stored in the storage **112**. Alternatively, the controller **111** may include a general-purpose arithmetic unit or a dedicated arithmetic unit. General-purpose arithmetic units and dedicated arithmetic units include integrated circuits. The integrated circuit constitutes a logic circuit.
(38) The storage **112** stores the data and the computer program. The data includes recipe data. The recipe data indicates a plurality of recipes. Each of the plurality of recipes defines the processing content and processing procedure for the substrates W. The storage **112** includes main memory. The main memory is, for example semiconductor memory. Examples of the main memory include read-only memory (ROM) and random access memory (RAM). The storage **112** may further include an auxiliary storage device. Examples of the auxiliary storage device include semiconductor memory and a hard disk drive. The storage **112** may include removable media.
(39) The configuration of the substrate processing apparatus **100** according to the present embodiment will be further described with reference to FIG. **2**. As illustrated in FIG. **2**, the substrate processing apparatus **100** further includes a phosphoric acid replenishment mechanism **4**, a diluting liquid supply mechanism **5**, a bubbling section **7**, an etching solution circulating section **8**, and an automatic cover **21**.
(40) The automatic cover **21** opens and closes the upper opening of the processing tank **3**. Specifically, the automatic cover **21** opens and closes the upper opening of the inner tank **31** and the upper opening of the outer tank **32**. In the present embodiment, the automatic cover **21** includes a first cover piece **22** and a second cover piece **23**. The first cover piece **22** and the second cover

piece **23** are free to open and close the upper opening of the processing tank **3**. The first cover piece **22** and the second cover piece **23** open and close, whereby the automatic cover **21** opens and closes like hinged double doors.
(41) Specifically, the first cover piece **22** is rotatable about a first rotation shaft P**1**. The first rotation axis P**1** extends in the Y-direction. The first rotation shaft P**1** supports the end portion of the first cover piece **22** on the opposite side to the center side of the automatic cover **21**. The second cover piece **22** is rotatable about a second rotation shaft P**2**. The second rotation axis P**2** extends in the Y-direction. The second rotation shaft P**2** supports the end portion of the second cover piece **23** on the opposite side to the center side of the automatic cover **21**.
(42) The control device **110** (controller **111**) brings the automatic cover **21** to an open state when making the substrate holding section **130** move from the retracted position (position in FIG. **1**A) to the processing position (position in FIG. **1**B). When the automatic cover **21** becomes in the open state, the upper opening of the processing tank **3** is opened, which allows the substrate W to be placed in the processing tank **3** (inner tank **31**). The control device **110** (controller **111**) brings the automatic cover **21** to a closed state during an etching process on the substrates W. When the automatic cover **21** is brought to the closed state, the upper opening of the processing tank **3** is closed. As a result, the inside of the processing tank **3** becomes a closed space.
(43) The control device **110** (controller **111**) brings the automatic cover **21** to an open state when making the substrate holding section **130** move from the processing position (position in FIG. **1**B) to the retracted position (position in FIG. **1**A). When the automatic cover **21** becomes in the open state, the upper opening of the processing tank **3** is opened, which allows the substrates W to be pulled up from the processing tank **3** (inner tank **31**).
(44) The phosphoric acid replenishment mechanism **4** will be then described with reference to FIG. **2**. The phosphoric acid replenishment mechanism **4** replenishes the etching solution E in the processing tank **3** with phosphoric acid. In the present embodiment, the phosphoric acid with which is replenished by the phosphoric acid replenishment mechanism **4** is a new liquid. Therefore, the phosphoric acid with which is replenished by the phosphoric acid replenishment mechanism **4** contains no silicon.
(45) The control device **110** (controller **111**) controls the phosphoric acid replenishment mechanism **4** during an etching process on the substrates W so that the etching solution E in the processing tank **3** is replenished with phosphoric acid, thereby causing a silicon concentration C in the etching solution E to vary. Specifically, during the etching process on the substrates W, the controller **111** causes the silicon concentration C in the etching solution E to vary from a first silicon concentration C**1** to a second silicon concentration C**2**. Here, the first silicon concentration C**1** indicates the silicon concentration C at the start of the etching process. The second silicon concentration C**2** may be a concentration different from the first silicon concentration C**1**. For example, the second silicon concentration C**2** is larger than the first silicon concentration C**1**.
(46) Specifically, when the substrates W are immersed in the etching solution E, the silicon nitride films Mb (see FIG. **7**) included in each substrate W react with the phosphoric acid to be etched. At this time, silicon is produced as a reactant. The generated silicon dissolves in the etching solution E. Therefore, when the etching solution E is replenished with no phosphoric acid during the etching process on the substrates W, the silicon concentration C in the etching solution E increases at a constant rate until the silicon nitride films Mb are removed. On the other hand, in the present embodiment, the phosphoric acid replenishment mechanism **4** appropriately replenishes the etching solution E with phosphoric acid during the etching process on the substrates W, thereby controlling the silicon concentration C in the etching solution E.
(47) In the present embodiment, the phosphoric acid replenishment mechanism **4** includes a phosphoric acid supply nozzle **41** and a phosphoric acid supply pipe **42**. The phosphoric acid supply nozzle **41** directs the stream of phosphoric acid to the processing tank **3**. The phosphoric acid supply pipe **42** allows the phosphoric acid to flow through to the phosphoric acid supply

nozzle **41**. The phosphoric acid supply nozzle **41** is an example of a phosphoric acid supply section.
(48) More specifically, the phosphoric acid supply nozzle **41** is placed above the processing tank **3**. The phosphoric acid supply nozzle **41** is a hollow tubular member. A plurality of ejecting holes are formed in the phosphoric acid supply nozzle **41**. In the present embodiment, the phosphoric acid supply nozzle **41** extends in the Y-direction. The plurality of ejecting holes in the phosphoric acid supply nozzle **41** are formed at equal intervals in the Y-direction. When phosphoric acid is supplied to the phosphoric acid supply nozzle **41** via the phosphoric acid supply pipe **42**, the phosphoric acid is ejected from the plurality of ejecting holes in the phosphoric acid supply nozzle **41** toward the processing tank **3**. The processing tank **3** is consequently supplied with the phosphoric acid. In the present embodiment, the phosphoric acid supply nozzle **41** is placed above the outer tank **32**. Therefore, phosphoric acid is ejected from the phosphoric acid supply nozzle **41** toward the outer tank **32**, and supplied to the outer tank **32**.
(49) The diluting liquid supply mechanism **5** will be then described with reference to FIG. **2**. The diluting liquid supply mechanism **5** supplies a diluting liquid to the processing tank **3**. The diluting liquid is consequently supplied to the etching solution E in the processing tank **3**. Specifically, the diluting liquid supply mechanism **5** includes a diluting liquid supply nozzle **51** and a diluting liquid supply pipe **52**.
(50) The diluting liquid supply nozzle **51** is placed above the processing tank **3**. The diluting liquid supply nozzle **51** is a hollow tubular member. A plurality of ejecting holes are formed in the diluting liquid supply nozzle **51**. In the present embodiment, the diluting liquid supply nozzle **51** extends in the Y-direction. The plurality of ejecting holes in the diluting liquid supply nozzle **51** are formed at equal intervals in the Y-direction.
(51) The diluting liquid supply pipe **52** allows the diluting liquid to flow through to the diluting liquid supply nozzle **51**. The diluting liquid is supplied to the diluting liquid supply nozzle **51** via the diluting liquid supply pipe **52**, and then ejected from the plurality of ejecting holes in the diluting liquid supply nozzle **51**.
(52) The etching solution E is heated. The etching solution E has, for example a temperature in 120° C. or higher and 160° C. or lower. Therefore, the water contained in the etching solution E evaporates. A diluting liquid is appropriately supplied to the etching solution E to cause a concentration value or a specific gravity value of the phosphoric acid in the etching solution E to maintain a target value.
(53) The bubbling section **7** will be then described with reference to FIG. **2**. The bubbling section **7** supplies air bubbles toward the plurality of substrates W immersed in the etching solution E in the inner tank **31**. Specifically, the bubbling section **7** includes a plurality of gas supply nozzles **71** and a gas supply pipe **72**. Note that although the bubbling section **7** includes two gas supply nozzles **71** in the present embodiment, the bubbling section **7** may include one gas supply nozzle **71** or three or more gas supply nozzles **71**.
(54) The plurality of gas supply nozzles **71** are placed on the bottom side of the inner tank **31**. More specifically, the plurality of gas supply nozzles **71** are arranged in the inner tank **31** in a location below the plurality of substrates W immersed in the etching solution E in the inner tank **31**.
(55) Each of the gas supply nozzles **71** is a hollow tubular member. A plurality of ejecting holes are formed in each of the gas supply nozzles **71**. In the present embodiment, the gas supply nozzles **71** extend in the Y-direction. The plurality of ejecting holes in each gas supply nozzle **71** are formed at equal intervals in the Y-direction.
(56) Gas is ejected from the ejecting holes in each of the gas supply nozzles **71**, whereby air bubbles are supplied toward the plurality of substrates W immersed in the etching solution E in the inner tank **31**. The gas is, for example an inert gas. Specifically, the gas may be nitrogen.
(57) The gas supply pipe **72** allows the gas to flow through to the plurality of gas supply nozzles **71**. The gas flows through the gas supply pipe **72**, so that air bubbles are supplied toward the plurality of substrates W immersed in the etching solution E in the inner tank **31**. As a result, the

non-uniformity of the silicon concentration C in the etching solution E is suppressed, thereby enabling uniform etching on the substrates W.

(58) The etching solution circulating section **8** will be then described with reference to FIG. **2**. The etching solution circulating section **8** circulates the etching solution E between the outer tank **32** and the inner tank **31**. Specifically, the etching solution circulating section **8** includes a plurality of etching solution supply nozzles **81**, a circulation pipe **82**, a circulation pump **83**, a circulation heater **84**, and a circulation filter **85**. Note that although the etching solution circulating section **8** includes two etching solution supply nozzles **81** in the present embodiment, the etching solution circulating section **8** may include one or three or more etching solution supply nozzles **81**.

(59) The plurality of etching solution supply nozzles **81** are arranged on the bottom side of the inner tank **31**. Each of the etching solution supply nozzles **81** is a hollow tubular member. A plurality of ejecting holes are formed in each of the etching solution supply nozzles **81**. In the present embodiment, the etching solution supply nozzles **81** extend in the Y-direction. The plurality of ejection holes in each of the etching solution supply nozzles **81** are formed at equal intervals in the Y-direction.

(60) An end of the circulation pipe **82** is connected to the outer tank **32**, and allows the etching solution E to flow in from the outer tank **32**. The circulation pipe **82** allows the etching solution E to flow through to the etching solution supply nozzles **81**.

(61) The circulation pump **83** is intervened in the circulation pipe **82**. The circulation pump **83** drives the etching solution E to flow through the circulation pipe **82** by the pressure of the fluid. As a result, the etching solution E flows from the outer tank **32** to the inner tank **31** via the circulation pipe **82**. Specifically, the etching solution E flows through the circulation pipe **82** to be ejected in the inner tank **31** from the ejecting holes in each of the etching solution supply nozzles **8**. That is, the etching solution E is supplied from the etching solution supply nozzles **81** in the inner tank **31**. The etching solution E is also ejected from the etching solution supply nozzles **81** in the inner tank **31**, whereby the etching solution E flows from the inner tank **31** toward the outer tank **32** through the upper end faces of the side walls of the inner tank **31**.

(62) The circulation heater **84** and the circulation filter **85** are intervened in the circulation pump **82**. The circulation heater **84** heats the etching solution E flowing through the circulation pipe **82**. Specifically, the circulation heater **84** heats the etching solution E at a temperature of 120° C. or higher and 160° C. or lower. The circulation filter **85** removes foreign bodies from the etching solution E flowing through the circulation pipe **82**.

(63) The configuration of the phosphoric acid replenishment mechanism **4** will be then described with reference to FIG. **3**. FIG. **3** is a diagram illustrating the phosphoric acid replenishment mechanism **4** included in the substrate processing apparatus **100** according to the present embodiment. As illustrated in FIG. **3**, the phosphoric acid replenishment mechanism **4** further includes an on-off valve **43**. The on-off valve **43** is intervened in the phosphoric acid supply pipe **42**.

(64) The on-off valve **43** is, for example a solenoid valve. The on-off valve **43** opens and closes the flow path of the phosphoric acid supply pipe **42** to control the flow of phosphoric acid flowing through the phosphoric acid supply pipe **42**. Specifically, when the on-off valve **43** is opened, phosphoric acid flows to the phosphoric acid supply nozzle **41** via the phosphoric acid supply pipe **42**. The phosphoric acid is consequently ejected from the phosphoric acid supply nozzle **41**. On the other hand, when the on-off valve **43** is closed, the flow of phosphoric acid is cut off, and the phosphoric acid supply nozzle **41** stops ejecting the phosphoric acid.

(65) The on-off valve **43** is controlled by the control device **110** (controller **111**). The controller **111** opens and closes the on-off valve **43** during the etching process on the substrates W to cause the silicon concentration C in the etching solution E to vary. Specifically, the controller **111** opens and closes the on-off valve **43** during the etching process on the substrates W to control a phosphoric acid replenishment flow rate F that is the flow rate of phosphoric acid that the etching solution E is

replenished with. As a result, the silicon concentration C in the etching solution E is varied.
(66) Specifically, the phosphoric acid replenishment mechanism **4** further includes a flowmeter **44**. The flowmeter **44** is intervened in the phosphoric acid supply pipe **42**. The flowmeter **44** measures the flow rate of phosphoric acid flowing through the phosphoric acid supply pipe **42**. The flowmeter **44** outputs a signal representing the measurement result to the controller **111**. The flowmeter **44** may be, for example an integrated flow meter.
(67) The storage **112** stores a set value for the phosphoric acid replenishment flow rate F. The controller **111** controls the opening and closing of the on-off valve **43** based on the phosphoric acid flow rate measured through the flowmeter **44** and the set value for the phosphoric acid replenishment flow rate F. Thus, the flow rate of phosphoric acid (phosphoric acid replenishment flow rate F) which the etching solution E is to be replenished with is controlled based on the flow rate of phosphoric acid measured through the flowmeter **44** and the set value for the phosphoric acid replenishment flow rate F.
(68) The substrate processing method according to the present embodiment will be then described with reference to FIG. **4**. FIG. **4** is a flow chart illustrating the substrate processing method according to the present embodiment. The substrate processing method according to the present embodiment may be carried out by the substrate processing apparatus **100** described with reference to FIGS. **1**A to **3**. The following describes the substrate processing method carried out by the substrate processing apparatus **100** described with reference to FIGS. **1**A to **3**. As illustrated in FIG. **4**, the substrate processing method according to the present embodiment includes Steps S**1** to S**3**.
(69) First, at the start of an etching process on a plurality of substrates W, the substrates W are immersed in the etching solution E (Step S**1**). Specifically, the substrate holding section **130** moves to the processing position. The plurality of substrates W held by the substrate holding section **130** are consequently placed in the inner tank **31** to be immersed in the etching solution E in the inner tank **31**. At this time, the silicon concentration C in the etching solution E stored in the processing tank **3** is the first silicon concentration C**1**.
(70) When the plurality of substrates W are immersed in the etching solution E, the etching process is performed on each substrate W with the etching solution E. The controller **111** causes the silicon concentration C in the etching solution E to vary during the etching process on the substrates W. Specifically, the controller **111** controls the on-off valve **43** of the phosphoric acid replenishment mechanism **4** to replenish the etching solution E in the processing tank **3** with phosphoric acid, thereby causing the silicon concentration C in the etching solution E to vary (Step S**2**). More specifically, the controller **111** causes the silicon concentration C in the etching solution E to vary from the first silicon concentration C**1** to the second silicon concentration C**2** by replenishing with phosphoric acid.
(71) When a predetermined time elapses after the plurality of substrates W are immersed in the etching solution E, the plurality of substrates W are pulled up from the etching solution E (Step S**3**), and the etching process in FIG. **4** is completed. Specifically, the substrate holding section **130** moves from the processing position to the retracted position. As a result, the plurality of substrates W held by the substrate holding section **130** are pulled up from the etching solution E in the inner tank **31**.
(72) Varying examples of the silicon concentration C will be then described with reference to FIGS. **1**A to **6**. FIG. **5** is a diagram illustrating a varying example of the silicon concentration C during an etching process by the substrate processing apparatus **100** according to the present embodiment. FIG. **6** is a diagram illustrating an example of a transition of the phosphoric acid replenishment flow rate F during the etching process by the substrate processing apparatus **100** according to the present embodiment.
(73) In FIG. **5**, the vertical axis indicates the silicon concentration C. The horizontal axis indicates the processing time t. Further, Graph CP**1** (solid line) illustrates a varying example of the silicon concentration C when the phosphoric acid replenishment mechanism **4** replenishes the etching

solution E with phosphoric acid. Graph CP**2** (broken line) illustrates a varying example of the silicon concentration C when the etching solution E is replenished with no phosphoric acid. In FIG. **6**, the vertical axis indicates the phosphoric acid replenishment flow rate F. The horizontal axis indicates the processing time t. Further, Graph FP illustrates an example of a transition of the flow rate of phosphoric acid, (phosphoric acid replenishment flow rate F), with which the phosphoric acid replenishment mechanism **4** replenishes the etching solution E.
(74) As illustrated in FIG. **5**, the silicon concentration C at the etching start time ts is the first silicon concentration C**1**, and the silicon concentration C at the etching end time to is the second silicon concentration C**2**. In the example illustrated in FIG. **5**, the first silicon concentration C**1** is a low concentration and the second silicon concentration C**2** is a high concentration. For example, the low concentration is 40 ppm or more and 50 ppm or less when the temperature of the etching solution E is 160° C. The high concentration is 60 ppm when the etching solution E is 160° C.
(75) After the etching process is started and the plurality of substrates W are immersed in the etching solution E, the silicon concentration C increases. As illustrated in Graph CP**2**, when the etching solution E is replenished with no phosphoric acid during the etching process on the substrates W, the silicon concentration C in the etching solution E increases sharply. On the other hand, in the present embodiment, the phosphoric acid replenishment mechanism **4** replenishes the etching solution E with phosphoric acid during the etching process on the substrates W to dilute the silicon concentration C with the new phosphoric acid. As a result, the silicon concentration C gradually increases as illustrated in Graph CP**1**.
(76) For example, as illustrated in Graph FP in FIG. **6**, the controller **111** starts an etching process to drive the on-off valve of the phosphoric acid replenishment mechanism **4** so that the phosphoric acid replenishment flow rate F maintains the first replenishment flow rate F**1** (constant value). As a result, the silicon concentration C gradually increases as illustrated in FIG. **5**. Note that the first replenishment flow rate F**1** indicates the phosphoric acid replenishment flow rate F at the start of the etching process on the substrates W. In the example in FIG. **6**, the first replenishment flow rate F**1** is set to a flow rate that gradually increases the silicon concentration C from a low concentration to a high concentration.
(77) As illustrated in FIG. **5**, when the silicon concentration C increases to just before the second silicon concentration C**2**, the controller **111** drives the on-off valve **43** of the phosphoric acid replenishment mechanism **4** so that the phosphoric acid replenishment flow rate F increases as illustrated in FIG. **6**. As a result, the silicon concentration C increases more gradually before the second silicon concentration C**2**, as illustrated in FIG. **5**.
(78) As illustrated in FIG. **5**, when the processing time t reaches the time t**1** before the etching end time te, the silicon concentration C reaches the second silicon concentration C**2**. At this time, the phosphoric acid replenishment flow rate F has increased to the second replenishment flow rate F**2** as illustrated in FIG. **6**. After the phosphoric acid replenishment flow rate F reaches the second replenishment flow rate F**2**, the controller **111** drives the on-off valve **43** of the phosphoric acid replenishment mechanism **4** so that the phosphoric acid replenishment flow rate F maintains the second replenishment flow rate F**2**. The silicon concentration C consequently maintains the second silicon concentration C**2** as illustrated in FIG. **5**. The second replenishment flow rate F**2** is a flow rate that causes the silicon concentration C to maintain a constant value.
(79) The substrates W that undergoes an etching process through the substrate processing apparatus **100** according to the present embodiment will be then described with reference to FIG. **7**. FIG. **7** is a diagram illustrating a substrate W before undergoing the etching process through the substrate processing apparatus **100** according to the present embodiment. Substrates W that undergo the etching process through the substrate processing apparatus **100** according to the present embodiment are used for three-dimensional flash memory (e.g., three-dimensional NAND flash memory), for example.
(80) In FIG. **7**, the substrate W includes a base material S and a multilayer structure M. The base

material S is in the shape of a thin film spreading in the XZ plane. The base material S is made of, for example silicon. The multilayer structure M is formed on the upper surface of the base material S. The multilayer structure M is formed so as to extend in the Y-direction from the upper surface of the base material S. The multilayer structure M includes silicon oxide films Ma and silicon nitride films Mb alternately stacked in the Y-direction. Each of the silicon oxide films Ma spreads parallel to the upper surface of the base material S. Each of the silicon nitride films Mb spreads parallel to the upper surface of the base material S.
(81) The multilayer structure M includes one cavity RE or more. The cavity RE reaches the base material S from the upper surface of the multilayer structure M, and part of the upper surface of the base material S is exposed through the cavity RE. Further, the side walls of the silicon oxide films Ma and the silicon nitride films Mb are exposed from the inner surface of the cavity RE. The cavity RE functions as a trench or a hole when the substrates W are used for semiconductor products, for example
(82) An etching process through the substrate processing apparatus **100** according to the present embodiment will be then described with reference to FIGS. **7** and **8**. FIG. **8** is a diagram illustrating an example of a substrate W after undergoing an etching process through the substrate processing apparatus **100** according to the present embodiment.
(83) When the substrate W is immersed in the etching solution E, the etching solution E enters the cavity RE. The etching solution E consequently comes into contact with the silicon oxide films Ma and the silicon nitride films Mb at the interface of the cavity RE.
(84) Etch rates (selectivity) of the silicon oxide films Ma and the silicon nitride films Mb by the etching solution E can be controlled by the silicon concentration C in the etching solution E under the condition that the temperature of the etching solution E and the specific gravity value (concentration) of the phosphoric acid are constant. Specifically, when the silicon concentration C in the etching solution E is a high concentration, an etch amount of the silicon oxide films Ma is sufficiently small, and substantially only the silicon nitride films Mb are etched in the multilayer structure M. On the other hand, when the silicon concentration C in the etching solution E is lower than the high concentration, the silicon oxide films Ma are etched together with the silicon nitride films Mb. The etch amount of the silicon oxide films Ma decreases as the silicon concentration C increases. Note that an etch amount of the silicon nitride films Mb is substantially constant and not affected by the silicon concentration C.
(85) In the example described with reference to FIGS. **5** and **6**, when the silicon concentration C is varied to the low concentration or the high concentration, the silicon concentration C is the low concentration in the initial stage of the etching process. The silicon oxide films Ma are therefore etched together with silicon nitride films Mb. Specifically, the silicon oxide films Ma and the silicon nitride films Mb are sequentially etched from their respective parts that are on the cavity RE side and in contact with the etching solution E. However, since the etch rate on the silicon nitride films Mb is higher than that on the silicon oxide films Ma, the etch amount of the silicon oxide films Ma is smaller than the etch amount of the silicon nitride films Mb.
(86) Then, the silicon concentration C gradually increases, whereby of the silicon oxide films Ma, their respective parts farther from the cavity RE come into contact with the etching solution E having a higher silicon concentration C. As a result, the silicon oxide films Ma have a smaller width in the Y-direction on the cavity RE side and a larger width in the Y-direction on the farther side from the cavity RE, as illustrated in FIG. **8**. Therefore, in the substrates W after the etching process, the gap G between the silicon oxide films Ma adjacent to each other in the stacked direction becomes wider as it is closer to the cavity RE and narrower as it is farther from the cavity RE.
(87) The configuration of the control device **110** will be then described with reference to FIG. **9**. FIG. **9** is a block diagram illustrating the configuration of the control device **110** included in the substrate processing apparatus **100** according to the present embodiment. As illustrated in FIG. **9**,

the control device **110** further includes an input section **113**.
(88) The input section **113** allows an operator to enter data through. The input section **113** is a user interface device for an operation by the operator. The input section **113** enters data into the controller **111** according to the operation by the operator. The controller **111** stores the data entered through the input section **113** in the storage **112**. The input section **113** includes, for example, a keyboard and a mouse. The input section **113** may include a touch sensor.
(89) A set value for the phosphoric acid replenishment flow rate F is entered through the input unit **113**. Specifically, the operator operates the input section **113** to enter a set value according to the structure of the semiconductor device to be manufactured by using substrates W after being processed through the substrate processing apparatus **100**. For example, the operator enters a set value for the phosphoric acid replenishment flow rate F according to the size of the gap G described with reference to FIG. **8**.
(90) More specifically, the operator enters, as the set value for the phosphoric acid replenishment flow rate F, data indicating the relationship between the processing time t of the etching process and the phosphoric acid replenishment flow rate F. For example, the operator may enter data corresponding to Graph FP described with reference to FIG. **6** as the set value for the phosphoric acid replenishment flow rate F. Specifically, the operator enters, as the set value for the phosphoric acid replenishment flow rate F, data indicating the first replenishment flow rate F**1**, data indicating the second replenishment flow rate F**2**, and data indicating varying timing from the first replenishment flow rate F**1** to the second replenishment flow rate F**2**. The controller **111** controls the on-off valve **43** of the phosphoric acid replenishment mechanism **4** based on a phosphoric acid flow rate measured through the flowmeter **44** and the set value for the phosphoric acid replenishment flow rate F. The phosphoric acid replenishment flow rate F consequently transitions as illustrated in Graph FP in FIG. **6**. The silicon concentration C in the etching solution E therefore varies as illustrated in Graph CP**1** in FIG. **5**, and the shape of each substrate W becomes the shape as illustrated in FIG. **8**.
(91) First to third varying examples of corresponding silicon concentrations C will be then described with reference to FIGS. **1**A to **14**. FIG. **10** is a diagram illustrating the first to third varying examples of the corresponding silicon concentrations C during their respective etching processes through the substrate processing apparatus **100** according to the present embodiment. FIG. **11** is a diagram illustrating first to third transitional examples of corresponding phosphoric acid replenishment flow rates F during the respective etching processes through the substrate processing apparatus **100** according to the present embodiment. FIGS. **12** to **14** illustrate first to third examples of corresponding substrates W after undergoing their respective etching processes through the substrate processing apparatus **100** according to the present embodiment.
(92) In FIG. **10**, the vertical axis indicates the silicon concentration C. The horizontal axis indicates the processing time t. Further, Graphs CP**11** to CP**13** illustrate first to third varying examples of corresponding silicon concentrations C. In FIG. **11**, the vertical axis indicates the phosphoric acid replenishment flow rate F. The horizontal axis indicates the processing time t. Further, Graphs FP**1** to FP**3** illustrate first to third transitional examples of corresponding flow rates of phosphoric acid, (phosphoric acid replenishment flow rate F), with which the phosphoric acid replenishment mechanism **4** replenishes the etching solution E.
(93) As illustrated in FIG. **10**, the first to third examples (Graphs CP**11** to CP**13**) differ from each other in the second silicon concentrations C**2** (C**21** to C**23**). Specifically, the second silicon concentration C**21** in the first example (Graph CP**11**) is higher than the second silicon concentration C**22** in the second example (Graph CP**12**). The second silicon concentration C**22** in the second example (Graph CP**12**) is higher than the second silicon concentration C**23** in the third example (Graph CP**13**). As illustrated in FIG. **11**, the first to third examples (Graphs FP**1** to FP**3**) differ from each other in the first replenishment flow rates F**1** (F**11** to F**13**). Specifically, the first replenishment flow rate F**11** in the first example (Graph FP**1**) is smaller than the first replenishment

flow rate F**12** in the second example (Graph FP**2**). The first replenishment flow rate F**12** in the second example (Graph FP**2**) is smaller than the first replenishment flow rate F**13** in the third example (Graph FP**3**).

(94) As illustrated in FIGS. **10** and **11**, the larger the first replenishment flow rate F**1** becomes, the smaller the second silicon concentration C**2** becomes. Further, as illustrated in FIGS. **12** to **14**, the smaller the second silicon concentration C**2** becomes, the wider the gap G becomes. Specifically, the smaller the second silicon concentration C**2** becomes, the larger the etching amount of the silicon oxide films Ma becomes. Therefore, the etching amount of the silicon oxide films Ma increases toward the cavity RE side where the contact time with the etching solution E becomes longer. The gap G becomes wider toward the cavity RE side. For example, the gap G of the substrate W in the first example of FIG. **12** is narrower on the cavity RE side than that of the substrate W in the second example of FIG. **13**. The gap G of the substrate W in the second example of FIG. **13** is narrower on the cavity RE side than that of the substrate W in the third example of FIG. **14**.

(95) A fourth varying example of the silicon concentration C will be then described with reference to FIGS. **15** and **16**. FIG. **15** is a diagram illustrating a fourth varying example of the silicon concentration C during an etching process through the substrate processing apparatus **100** according to the present embodiment. FIG. **16** is a diagram illustrating a fourth example of a substrate W after undergoing the etching process through the substrate processing apparatus **100** according to the present embodiment.

(96) In FIG. **15**, the vertical axis indicates the silicon concentration C. The horizontal axis indicates the processing time t. Further, Graph CP**14** illustrates a fourth varying example of the silicon concentration C. As illustrated in FIG. **15**, in the fourth example, the first silicon concentration C**1** is higher than the second silicon concentration C**2**. Specifically, the first silicon concentration C**1** is a high concentration, and the second silicon concentration C**2** is a low concentration. In this case, since the silicon concentration C is the high concentration in the initial stage of the etching process, almost only the silicon nitride films Mb are etched in the multilayer structures M. Then, the silicon concentration C gradually decreases. The etching amount of the silicon oxide films Ma then becomes larger toward the cavity RE side where the contact time with the etching solution E becomes longer as illustrated in FIG. **16**. As a result, the width of each silicon oxide film Ma in the Y-direction becomes small on the cavity RE side, and increases as the distance from the cavity RE increases.

(97) The first embodiment has been described above with reference to FIGS. **1**A to **16**. The present embodiment enables the silicon concentration C in the etching solution E to vary during the etching process on the substrates W. The present embodiment therefore makes it possible to control the shape of the silicon oxide films Ma to form the substrates W into a special shape.

Second Embodiment

(98) The second embodiment will be then described with reference to FIG. **17**. However, matters different from those of the first embodiment will be described, and description of the same matters as those of the first embodiment will be omitted. The second embodiment is different from the first embodiment in that a substrate processing apparatus **100** includes a silicon concentration measuring device **86**.

(99) FIG. **17** is a cross-sectional view illustrating the configuration of the substrate processing apparatus **100** according to the present embodiment. As illustrated in FIG. **17**, the substrate processing apparatus **100** according to the present embodiment further includes the silicon concentration measuring device **86**. The silicon concentration measuring device **86** measures the silicon concentration C in the etching solution E. The silicon concentration measuring device **86** outputs a signal representing the measurement result to a controller **111**.

(100) In the present embodiment, the silicon concentration measuring device **86** is intervened in a circulation pipe **82**. The silicon concentration measuring device **86** therefore measures the silicon

concentration C in the etching solution E flowing through the circulation pipe **82**. Specifically, the silicon concentration measuring device **86** is placed on the downstream side (inner tank **31** side) of a circulation filter **85**. The silicon concentration measuring device **86** therefore measures the silicon concentration C in the etching solution E after foreign bodies are removed. This makes it possible to enhance the precision of the measurement result of the silicon concentration C in the etching solution E through the silicon concentration measuring device **86**.
(101) In the present embodiment, the controller **111** controls the opening and closing of an on-off valve **43** (see FIG. **3**) of a phosphoric acid replenishment mechanism **4** based on the silicon concentration C measured through the silicon concentration measuring device **86**. In other words, the controller **111** controls the flow rate of phosphoric acid, (phosphoric acid replenishment flow rate F), with which the phosphoric acid replenishment mechanism **4** replenishes the etching solution E based on the silicon concentration C measured through the silicon concentration measuring device **86**.
(102) More specifically, the controller **111** controls the phosphoric acid replenishment flow rate F based on a set value for the silicon concentration C stored in storage **112** and the silicon concentration C measured through the silicon concentration measuring device **86**.
(103) Specifically, the storage **112** stores, as a set value for the silicon concentration C, data indicating the relationship between the processing time t of the etching process and the silicon concentration C. Alternatively, the storage **112** stores data indicating the relationship between the processing time t of the etching process and the varying rate of the silicon concentration C. For example, the storage **112** may store the data corresponding to Graph CP**1** described with reference to FIG. **5**. In this case, the controller **111** controls the on-off valve **43** (see FIG. **3**) of the phosphoric acid replenishment mechanism **4** so that the silicon concentration C measured through the silicon concentration measuring device **86** follows the variation of the silicon concentration C as illustrated in Graph CP**1** of FIG. **5**. Each substrate W is formed into the shape as illustrated in FIG. **8**.
(104) The second embodiment has been described above with reference to FIG. **17**. The present embodiment enables the silicon concentration C in the etching solution E to vary during the etching process on the substrates W like the first embodiment. The present embodiment makes it possible to control the shape of the silicon oxide films Ma to form the substrates W into a special shape.
Third Embodiment
(105) A third embodiment will be described with reference to FIGS. **18** to **24**. However, the matters different from those of the first and second embodiments will be described, and the same matters as those of the first and second embodiments will be omitted. The third embodiment is different from the first and second embodiments in that a substrate processing apparatus **100** includes a silicon supply mechanism **45**.
(106) FIG. **18** is a cross-sectional view illustrating the configuration of the substrate processing apparatus **100** according to the present embodiment. As illustrated in FIG. **18**, the substrate processing apparatus **100** according to the present embodiment further includes the silicon supply mechanism **45**. The silicon supply mechanism **45** supplies a silicon containing liquid to the phosphoric acid that an etching liquid E is replenished with. The silicon containing liquid is a liquid that contains silicon. Note that the silicon containing liquid is, for example a suspension containing silicon. Hereinafter, the phosphoric acid that the etching liquid E is replenished with may be referred to as “supplemental phosphoric acid”.
(107) Specifically, the silicon supply mechanism **45** includes a silicon supply pipe **451**. An end of the silicon supply pipe **451** is connected to a phosphoric acid supply pipe **42**. The silicon supply pipe **451** allows the silicon containing liquid to flow through to the phosphoric acid supply pipe **42**. The silicon containing liquid is therefore supplied to the supplemental phosphoric acid flowing through the phosphoric acid supply pipe **42**. As a result, the supplemental phosphoric acid containing silicon is ejected from a phosphoric acid supply nozzle **41** toward an outer tank **32**. Hereinafter, the supplemental phosphoric acid containing silicon may be referred to as “silicon

containing phosphoric acid".
(108) The configuration of the silicon supply mechanism **45** will be then described with reference to FIG. **19**. FIG. **19** is a diagram illustrating a phosphoric acid replenishment mechanism **4** and the silicon supply mechanism **45** included in the substrate processing apparatus **100** according to the present embodiment. As illustrated in FIG. **19**, the silicon supply mechanism **45** further includes an on-off valve **452**. The on-off valve **452** is intervened in the silicon supply pipe **451**. Note that in the following description, the on-off valve **43** of the phosphoric acid replenishment mechanism **4** may be described as a "first on-off valve **43**", and the on-off valve **452** of the silicon supply mechanism **45** may be described as a "second on-off valve **452**".
(109) The second on-off valve **452** is, for example a solenoid valve. The second on-off valve **452** opens and closes the flow path of the silicon supply pipe **451** to control the flow of the silicon containing liquid flowing through the silicon supply pipe **451**. Specifically, when the second on-off valve **452** is opened, the silicon containing liquid flows to the phosphoric acid supply pipe **42** through the silicon supply pipe **451**. The silicon containing phosphoric acid is consequently ejected from the phosphoric acid supply nozzle **41**. On the other hand, when the second on off valve **452** is closed, the flow of the silicon containing liquid is cut off, and the supply of silicon (silicon containing liquid) to the phosphoric acid flowing through the phosphoric acid supply pipe **42** is stopped.
(110) The second on-off valve **452** is controlled by a control device **110** (controller **111**). The controller **111** opens and closes the first on-off valve **43** and the second on-off valve **452** during an etching process on substrates W to cause the silicon concentration C in an etching solution E to vary. Specifically, the controller **111** opens and closes the second on-off valve **452** during the etching process on the substrates W to control a silicon supply flow rate R that is a flow rate of the silicon containing liquid to be supplied to the supplemental phosphoric acid. The controller **111** also opens and closes the first on-off valve **43** during the etching process on the substrates W to control the flow rate of the silicon containing phosphoric acid and the flow rate of the supplemental phosphoric acid (new liquid).
(111) Specifically, the silicon supply mechanism **45** further comprises a flowmeter **453**. The flowmeter **453** is intervened in the silicon supply pipe **451**. The flow meter **453** measures the flow rate of the silicon containing liquid flowing through the silicon supply pipe **451**. The flowmeter **453** outputs a signal representing the measurement result to the controller **111**. The flowmeter **453** may be, for example an integrated flowmeter.
(112) Storage **112** stores a set value for the silicon supply flow rate R. The controller **111** controls the opening and closing of the second on-off valve **452** based on the flow rate of the silicon containing liquid measured through the flowmeter **453** and the set value for the silicon supply flow rate R. In other words, the controller **111** controls a supply flow rate (silicon supply flow rate R) of the silicon containing liquid to be supplied to the supplemental phosphoric acid based on the flow rate of the silicon containing liquid measured through the flowmeter **453** and the set value for the silicon supply flow rate R.
(113) First to third varying examples of corresponding silicon concentrations C will be then described with reference to FIGS. **18** to **24**. FIG. **20** is a diagram illustrating the first to third varying examples of the corresponding silicon concentrations C during their respective etching processes through the substrate processing apparatus **100** according to the present embodiment. FIG. **21** is a diagram illustrating first to third transitional examples of corresponding silicon supply flow rates R during the respective etching processes through the substrate processing apparatus **100** according to the present embodiment. FIGS. **22** to **24** illustrate first to third examples of corresponding substrates W after undergoing the respective etching processes through the substrate processing apparatus **100** according to the present embodiment.
(114) In FIG. **20**, the vertical axis indicates the silicon concentration C. The horizontal axis indicates the processing time t. Further, Graphs CP**21** to CP**23** illustrate first to third varying

examples of corresponding silicon concentrations C. In FIG. **21**, the vertical axis indicates the silicon supply flow rate R. The horizontal axis indicates the processing time t. Further, Graphs RP**1** to RP**3** illustrate first to third varying examples of corresponding supply flow rates (silicon supply flow rates R) of silicon containing liquids.

(115) As illustrated in FIG. **20**, the silicon concentration C varies from a first silicon concentration C**1** to a second silicon concentration C**2** during the etching process on the substrates W. In the example of FIG. **20**, the second silicon concentration C**2** has a higher concentration than the first silicon concentration C**1**.

(116) After the etching process is started and the plurality of substrates W are immersed in the etching solution E, the silicon concentration C increases. In the first to third examples, their respective lengths of time required for the silicon concentration C to vary from the first silicon concentration C**1** to the second silicon concentration C**2** are different from each other. Hereinafter, the length of time required for the silicon concentration C to vary from the first silicon concentration C**1** to the second silicon concentration C**2** may be referred to as a “silicon concentration varying period”.

(117) Specifically, a silicon concentration varying period (ts to t**11**) in the first example (Graph CP**21**) is shorter than a silicon concentration varying period (ts to t**21**) in the second example (Graph CP**22**). In other words, the silicon concentration C in the first example reaches the second silicon concentration C**2** from the first silicon concentration C**1** at a timing earlier than that in the second example. Therefore, a varying rate of the silicon concentration C in the first example is larger than that of the silicon concentration C in the second example.

(118) A silicon concentration varying period (ts to t**21**) in the second example (Graph CP**22**) is shorter than a silicon concentration varying period (ts to t**31**) in the third example (Graph CP**23**). In other words, the silicon concentration C in the second example reaches the second silicon concentration C**2** from the first silicon concentration C**1** at a timing earlier than that in the third example. Therefore, a varying rate of the silicon concentration C in the second example is larger than a varying rate of the silicon concentration C in the third example.

(119) As illustrated in FIG. **21**, the silicon containing liquid is supplied to the supplemental phosphoric acid at the start of the etching process (etching start time ts). The etching solution E is therefore replenished with the silicon containing phosphoric acid at the start of the etching process (etching start time ts). Specifically, when the etching process is started and the plurality of substrates W are immersed in the etching solution E, the silicon supply mechanism **45** supplies the silicon containing liquid to the supplemental phosphoric acid at a constant supply flow rate, and stops the supply of the silicon containing liquid after a predetermined time elapses.

(120) As illustrated in Graphs RP**1** to RP**3**, in the first to third examples, their respective silicon supply periods and silicon supply flow rates R are different from each other. Herein, each of the silicon supply periods is the length of time for supplying the silicon containing liquid.

(121) Specifically, a silicon supply period (ts to t**11**) in the first example (Graph RP**1**) is shorter than a silicon supply period (ts to t**21**) in the second example (Graph RP**2**), and the silicon supply flow rate R**1** in the first example (graph RP**1**) is larger than a silicon supply flow rate R**2** in the second example (Graph RP**2**). As a result, as described with reference to FIG. **20**, the silicon concentration C in the first example reaches the second silicon concentration C**2** from the first silicon concentration C**1** at a timing earlier than that in the second example, and a varying rate of the silicon concentration C in the first example is larger than a varying rate of the silicon concentration C in the second example.

(122) The silicon supply period (ts to t**21**) in the second example (Graph RP**2**) is shorter than a silicon supply period (ts to t**31**) in the third example (Graph RP**3**), and the silicon supply flow rate R**2** in the second example (Graph RP**2**) is larger than a silicon supply flow rate R**3** in the third example (Graph RP**3**). As a result, as described with reference to FIG. **20**, the silicon concentration C in the second example reaches the second silicon concentration C**2** from the first silicon

concentration C**1** at a timing earlier than that in the third example, and the varying rate of the silicon concentration C in the second example is larger than a varying rate of the silicon concentration C in the third example.
(123) After the end of the silicon supply period, a new phosphoric acid solution is supplied to the etching solution E from the phosphoric acid replenishment mechanism **4** in order to maintain the silicon concentration C at the second silicon concentration C**2**.
(124) As described with reference to FIGS. **20** and **21**, the silicon concentration C in the first example reaches the second silicon concentration C**2** from the first silicon concentration C**1** at a timing earlier than that in the second example. As a result, as illustrated in FIGS. **22** and **23**, a gap G in each substrate W (FIG. **22**) in the first example becomes narrower than a gap G of each substrate W (FIG. **23**) in the second example. Further, the silicon concentration C in the second example reaches the second silicon concentration C**2** from the first silicon concentration C**1** at a timing earlier than that in the third example. As a result, as illustrated in FIGS. **23** and **24**, the gap G of each substrate W (FIG. **23**) in the second example becomes narrower than a gap G of each substrate W (FIG. **24**) in the third example.
(125) Here, data entered in the control device **110** by an operator will be described with reference to FIG. **9**. In the present embodiment, the operator enters data indicating the relationship between the processing time t of the etching process and the silicon supply flow rate R as the set value for the silicon supply flow rate R. For example, the operator may enter data corresponding to Graph RP**1** described with reference to FIG. **21** as the set value for the silicon supply flow rate R. In this case, the controller **111** controls the silicon supply flow rate R as illustrated in Graph RP**1** of FIG. **21** based on the flow rate of the silicon containing liquid measured through the flowmeter **453** of the silicon supply mechanism **45** and the set value for the silicon supply flow rate R. As a result, the silicon concentration C in the etching solution E varies as illustrated in Graph CP**21** of FIG. **20**, and the shape of each substrate W becomes the shape as illustrated in FIG. **22**.
(126) The third embodiment has been described above with reference to FIGS. **18** to **24**. The present embodiment enables the silicon concentration C in the etching solution E to vary during the etching process on the substrates W, like the first embodiment. The present embodiment therefore makes it possible to control the shape of the silicon oxide films Ma to form each substrate W into a special shape.
(127) Note that the substrate processing apparatus **100** may include a silicon concentration measuring device **86** like the second embodiment. In this case, the controller **111** controls the opening and closing of the on-off valve **452** of the silicon supply mechanism **45** based on the silicon concentration C measured through the silicon concentration measuring device **86**. In other words, the controller **111** controls the flow rate of the silicon containing liquid (silicon supply flow rate R) to be supplied to the supplemental phosphoric acid based on the silicon concentration C measured through the silicon concentration measuring device **86**.
(128) More specifically, the controller **111** controls the silicon supply flow rate R based on the set value for the silicon concentration C stored in the storage **112** and the silicon concentration C measured through the silicon concentration measuring device **86**.
(129) Specifically, the storage **112** stores data indicating the relationship between the processing time t of the etching process and the silicon concentration C as the set value for the silicon concentration C. Alternatively, the storage **112** stores data indicating the relationship between the processing time t of the etching process and the varying rate of the silicon concentration C. For example, the storage **112** may store data corresponding to Graph CP**21** described with reference to FIG. **20**. In this case, the controller **111** controls the opening and closing of the on-off valve **452** of the silicon supply mechanism **45** so that the silicon concentration C measured through the silicon concentration measuring device **86** follows the variation of the silicon concentration C as illustrated in Graph CP**21** of FIG. **20**.
(130) The embodiments of the present disclosure have been described above with reference to the

drawings (FIGS. **1**A to **24**). However, the present disclosure is not limited to the above-described embodiments, and may be implemented in various aspects without departing from the gist thereof. In addition, the plurality of components disclosed in the above embodiments can be appropriately altered. For example, a component of all the components in an embodiment may be added as a component of another embodiment, or some components of all the components in an embodiment may be removed.

(131) The drawings schematically illustrate each component as a main body in order to make it easier to understand. The thickness, length, numbers, intervals and the like of illustrated components may differ from the actual ones for the convenience of the drawing. Further, each component illustrated in the above embodiments is an example and not particularly limited, and various modifications may be made without substantially deviating from the effects of the present disclosure.

(132) For example, in the embodiments described with reference to FIGS. **1**A to **24**, the etching solution E in the processing tank **3** is replenished with the supplemental phosphoric acid from the outside of the processing tank **3**. The etching solution E may however be supplied with the supplemental phosphoric acid inside the processing tank **3**.

Claims

1. A substrate processing method performing an etching process on a substrate with an etching solution in a processing tank, the substrate including silicon oxide films and silicon nitride films stacked alternately, the etching solution containing phosphoric acid, the substrate processing method including: immersing the substrate in the etching solution and start etching; and increasing silicon concentration of the etching solution from a first silicon concentration to a second silicon concentration that is higher than the first silicon concentration by replenishing the etching solution in the processing tank with phosphoric acid during the etching process on the substrate to cause the silicon concentration in the etching solution to increase first at a first rate, and then, before reaching the second silicon concentration, increasing a replenishment flow rate of the phosphoric acid to increase the silicon concentration at a second rate, the second rate being smaller than the first rate.
2. The substrate processing method according to claim 1, wherein in the replenishing, the replenishment flow rate of the phosphoric acid that the etching solution is replenished with is controlled based on a set value for a replenishment flow rate of phosphoric acid that is set according to a structure of a semiconductor device to be manufactured using the substrate.
3. The substrate processing method according to claim 2, wherein the structure of the semiconductor device indicates size of a gap between the silicon oxide films adjacent to each other in a stacking direction in the semiconductor device.
4. The substrate processing method according to claim 1, wherein in the replenishing, the replenishment flow rate of the phosphoric acid that the etching solution is replenished with is controlled based on the silicon concentration in the etching solution, the silicon concentration being measured during the etching process on the substrate.
5. The substrate processing method according to claim 1, wherein in the replenishing, the phosphoric acid that the etching solution is replenished with is supplied with a silicon containing liquid, the silicon containing liquid being a liquid that contains silicon.
6. The substrate processing method according to claim 5, wherein in the replenishing, a supply flow rate of the silicon containing liquid to be supplied to the phosphoric acid is controlled based on a set value for a supply flow rate of a silicon containing liquid that is set according to a structure of a semiconductor device to be manufactured using the substrate.
7. The substrate processing method according to claim 6, wherein the structure of the semiconductor device indicates size of a gap between the silicon oxide films adjacent to each other in a stacking direction in the semiconductor device.

8. The substrate processing method according to claim 5, wherein in the replenishing, a supply flow rate of the silicon containing liquid to be supplied to the phosphoric acid is controlled based on the silicon concentration in the etching solution measured during the etching process on the substrate.
9. The substrate processing method according to claim 1, wherein in the replenishing, the replenishment flow rate of the phosphoric acid that the etching solution is replenished with is controlled based on data stored in a storage and indicative of a predetermined silicon concentration variation such that the silicon concentration in the etching solution in the processing tank measured during the etching process on the substrate follows the predetermined silicon concentration variation.
10. The substrate processing method according to claim 9, wherein the data stored in the storage indicates a relationship between a processing time of the etching process and the silicon concentration.
11. The substrate processing method according to claim 9, wherein the data stored in the storage indicates a relationship between a processing time of the etching process and a varying rate of the silicon concentration.
12. The substrate processing method of claim 1, wherein the increasing the silicon concentration starts when etching starts.